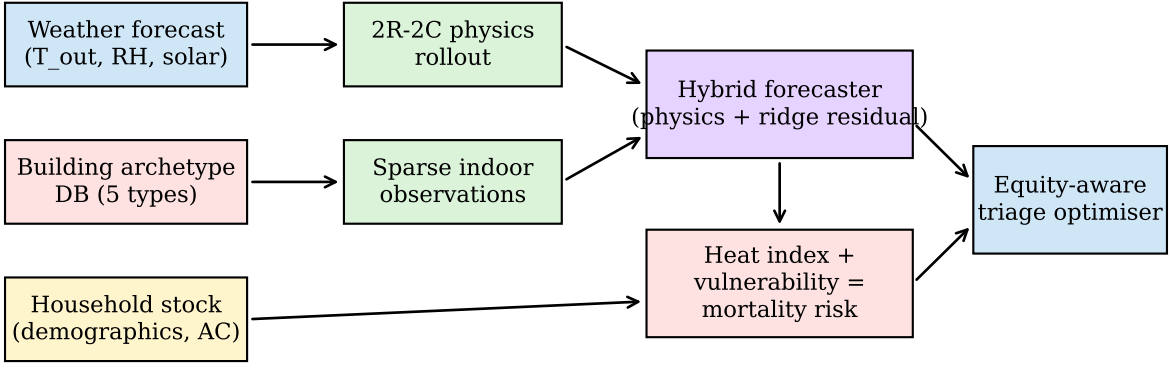ThermoGuard system architecture